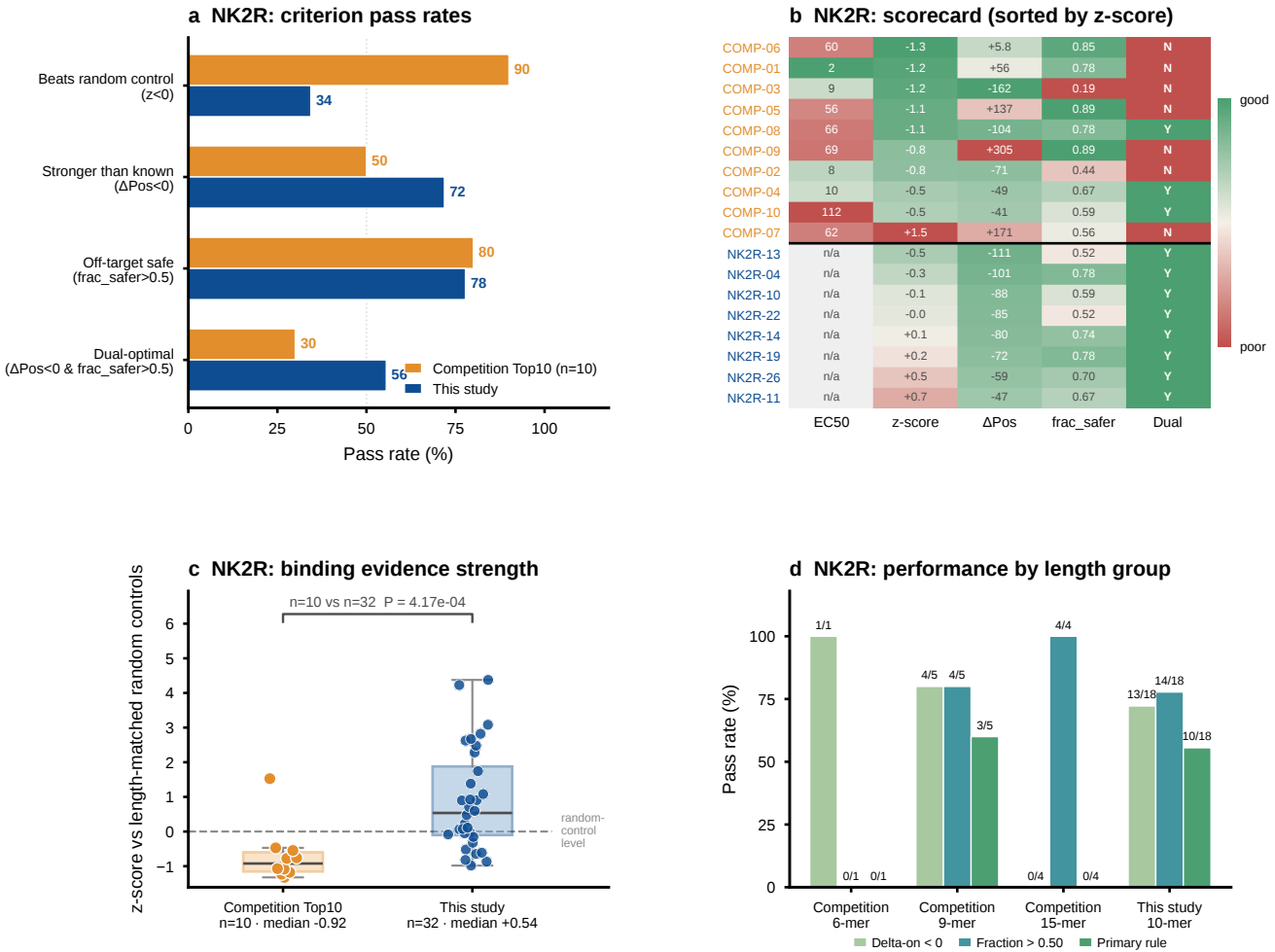
a NK2R: criterion pass rates
Beats random control (z<0)
Stronger than known (ΔPos<0)
Off-target safe (frac_safer>0.5)
Dual-optimal (ΔPos<0 & frac_safer>0.5)
90
34
50
72
80
78
30
56
Competition Top10 (n=10)
This study
0
25
50
75
100
Pass rate (%)
b NK2R: scorecard (sorted by z-score)
COMP-06 60 -1.3 +5.8 0.85 N
COMP-01 2 -1.2 +56 0.78 N
COMP-03 9 -1.2 -162 0.19 N
COMP-05 56 -1.1 +137 0.89 N
COMP-08 66 -1.1 -104 0.78 Y
COMP-09 69 -0.8 +305 0.89 N
COMP-02 8 -0.8 -71 0.44 N
COMP-04 10 -0.5 -49 0.67 Y
COMP-10 112 -0.5 -41 0.59 Y
COMP-07 62 +1.5 +171 0.56 N
NK2R-13 n/a -0.5 -111 0.52 Y
NK2R-04 n/a -0.3 -101 0.78 Y
NK2R-10 n/a -0.1 -88 0.59 Y
NK2R-22 n/a -0.0 -85 0.52 Y
NK2R-14 n/a +0.1 -80 0.74 Y
NK2R-19 n/a +0.2 -72 0.78 Y
NK2R-26 n/a +0.5 -59 0.70 Y
NK2R-11 n/a +0.7 -47 0.67 Y
EC50
z-score
ΔPos
frac_safer
Dual
good
poor
c NK2R: binding evidence strength
n=10 vs n=32 P = 4.17e-04
z-score vs length-matched random controls
6
5
4
3
2
1
0
−1
random-control level
Competition Top10
n=10 · median -0.92
This study
n=32 · median +0.54
d NK2R: performance by length group
Pass rate (%)
100
75
50
25
0
1/1
0/1
0/1
4/5
4/5
3/5
0/4
4/4
0/4
13/18
14/18
10/18
Competition 6-mer
Competition 9-mer
Competition 15-mer
This study 10-mer
Delta-on < 0
Fraction > 0.50
Primary rule